CREATING A UNIVERSAL INFORMATION LANGUAGE

OSCAR BENDER-STONE

ABSTRACT. Welkin is a formalized information language. We introduce its use cases and rigorously define its syntax and semantics. From there, we introduce the bootstrap, making Welkin completely self-contained. [TODO: determine how to phrase soundness and incompleteness. Should we include these?].

CONTENTS

1. Introduction . . . 1
2. Rationale . . . 5
   2.1. General Requirements . . . 5
   2.2. Units . . . 5
   2.3. Mechanizing Information . . . 6
   2.4. Base Operations . . . 7
   2.5. Bootstrapping . . . 8
3. Foundations . . . 9
   3.1. Base Rules . . . 9
   3.2. Base Recursor . . . 13
   3.3. Queries . . . 15
4. Information . . . 15
   4.1. Serial Soundness . . . 16
   4.2. Defining Information . . . 17
5. Syntax . . . 17
   5.1. Words . . . 17
   5.2. Terminals . . . 18
   5.3. EBNF Notation and Parse Trees . . . 20
   5.4. The Welkin Grammar . . . 20
   5.5. Proof of Unambiguity . . . 21
6. Organizing Information . . . 25
   6.1. Computable Measures and Size . . . 26
   6.2. Localized Size Compression . . . 26
   6.3. Query Compression . . . 26
7. Conclusion . . . 26
   7.1. Comparisons to Existing Literature . . . 27
   7.2. Programming Languages and Formal Verification . . . 27
   7.3. Mathematical and Scientific Knowledge . . . 27
   7.4. Humanities . . . 28
References . . . 28

DRAFT

1. INTRODUCTION

[TODO[MEDIUM]: determine if the phrase: “see Section ?” should be used, or “refer to” instead, to fully include all readers.]

Information Management is the study of systematically storing and organizing data. With the massive growth of data, particularly on the web, simply storing data to *use* it. Raw databases are increasingly structured into organized *information bases* to prove

---

This project used Google Gemini (https://gemini.google.com/) to aid in research and help fix formatting.

Date: March 5, 2026.

context surrounding the source and nature of data. The conceptual foundation for this process often traces back to Ackoff [Ack89]. His data-to-wisdom hierarchy posits that information is the process of transforming raw data into structured content. In turn, this content can be used to answer user queries. However, the term "information" has not been standardized in the literature and is used with distinct meanings. For example, Buckland [Buc91] recognizes three different ways information is interpreted: as a *process*, as *knowledge*, and as *a thing*. While each of these are important, Buckland emphasizes that to *use* information, one must work with information as a thing—a tangible record like a document or digital file. Another author, Floridi, builds upon this tangible idea by requiring data be truthful [Flo10]. Information must not only be in concrete records, but also accurately represent a topic of interest. As shown in this small sample of authors, information is a rich term, with an active pursuits to best manage it.

Despite these extensive ideas, applying Information Management in real world projects reveals ongoing problems. One of these issues is the *depth* of domains. For example, at major retailers like Amazon, inventory and recommendation systems use immense amounts of data to power personalized recommendations to users [SL17]. Another issue is the immense breadth of different subjects. As a well known example, Wikipedia struggles to constantly update its vast and interconnected web of knowledge [Mes+15]. Along with the depth and breadth of different disciplines, standardization is complex. Most systems have their own unique data formats, so making **interoperable** formats is hard, which is the ability to transfer one data format to another. For instance, transferring information between two medical platforms is notoriously difficult, even if they store the *same* information about a patient [Rei17]. These problems continue to be a challenge in effectively utilizing frameworks for information.

In consideration of these issues, several solutions partially resolve some of them, but not completely. The following provides a small list of major existing solutions. Reading these is optional and not required to understand this thesis. We will not revisit these until the conclusion, see Section 7.1.

- **Resource Description Framework (RDF) Triple Stores:** These systems serve as the primary implementations of an internet standard called the Web Ontology Language (OWL) [Hit+09]. This system stores information as triples $\text{subject} - \text{predicate} \rightarrow \text{object}$. From there, the language can enforce rules and ensure new relationships are derived *only* from from previous ones. OWL has been highly successful in providing a machine-readable format to websites across the internet [Hit21]. However, despite their widespread adoption, RDF stores struggle with extensive sources of data. Enforcing logical rules can be expensive [Zhu+23]. Additionally, because OWL treats missing sources of data as *unknown* rather than *false*, it often fails to resolve conflicts between two sources [Hit21].
- **Labeled Property Graphs (LPGs):** This is an architecture defined by its ability to store rich, internal metadata within its components. Building on a network, consisting of nodes and edges between them, LPGs enable both nodes (entities) and edges (relationships) to possess a set of key-value pairs known as *properties* [Ang18]. The internal metadata allows systems to efficiently traverse networks, as attributes can be filtered and weighed on a single query pass. Due to their operational efficiency, LPGs are the primary choice in large scale

commercial products, particularly real-time fraud detection tools and recommendation systems [RWE15]. Although LPGs are widely used, they are siloed by the lack of standards, making interoperability between databases challenging. Furthermore, LPGs lack an extensive logical engine, resulting in external tooling or complex query scripts [RWE15].

- **Cyc:** Designed to encompass broad subjects, Cyc is the largest proprietary database of knowledge, or *knowledge base*, known to date [Len95]. It achieves reliable queries by providing a logical engine, based on millions of hard coded "common sense" rules. However, despite its impressive scope, it relies on human experts to manually encode every rule. This limits how well the database scales [Har07]. Another issue is the absence of an active open standard, a result of the publicly available version, OpenCyc, being discontinued in 2017 [KBp].
- **fcaR (Formal Concept Analysis in R):** This tool is a modern implementation for the field of Formal Concept Analysis (FCA). FCA studies relationships through *formal concepts*. The central idea is the *concept lattice*, a hierarchical data structure that maps objects to attributes in a fixed domain. Given only a database, fcaR can automatically identify the underlying core concepts and derived concepts [MDV26]. While being highly effective in specialized domains, concept lattices can become impractically large as the number of attributes increases [HS13]. This makes it difficult to interpret results when applied to large-scale databases or when bridging disparate domains [HS13].

This thesis introduces a language to resolve these issues for Information Management. I call this language **Welkin**, based on an old German word meaning cloud [Dic25]. This word represents the expansive nature of the language.

The core result of this thesis is proving that Welkin fulfills three goals: it is **universal**, **scalable**, and **standardized**. For details, see Table 1. The fundamental building block is based on **representations**, containing a relationships between a **sign** representing a **sign** in a **context**.[1] Contexts provide flexibility to define domain terminology, and are provided rules inspired by McCarthy's work in AI systems [McC93]. Moreover, truth is defined relative to contexts and enables the user to recognize *under which conditions* a statement is true. In this way, the information base keeps conflicting claims contained, providing a clear separation from other contexts.

Building on representations, information is defined as a property preserved under equivalences relative to a context. To make the information base scalable, the base operations are kept as simple as possible, and a procedure to adaptively organize information. The simplicity of the language aids in standardization as well. For the highest reassurance, we create a restricted, finite subset of the language. This subset facilitates proofs in this thesis and enables ease of implementation.

[1]There are similarities with representations and Peirce's semiotics. This theory studies the relationship between a symbol, the object it represents, and the interpretation that provides that meaning [Atk23]. Our notion is different in that contexts are a generalized interpretant, see Section 2.2.

DRAFT

Goal	Name	Description
Goal 1	**Universality**	The language must enable any user created parameters, whose symbolic representation is accepted a computable function. Every computable function must be definable in the language.
Goal 2	**Scalability**	The database must appropriately scale to broad representations of information. Local queries must be efficient. Certificates must be available to prove cases where optimal representations have been achieved.
Goal 3	**Standardization**	The language needs a rigorous and formal specification. Moreover, the bootstrap must be formalized, as well as an abstract machine model. The grammar and bootstrap must be fixed to ensure complete forwards and backwards compatibility. Certificates must be reliably checked and rely on a low level of trust, or a small Trusted Computing Base.

TABLE 1. Goals for the Welkin language.

DRAFT

This thesis is organized according to Table 2.

Section Number	Title	Description
Section 2	**Rationale**	Introduces Welkin at a high level, with guiding examples.
Section 3	**Foundations**	Introduces the core theory behind Welkin, which is the starting point to specify the user-facing language.
Section 5	**Syntax**	Provides the grammar and proof that it is unambiguous.
Section 6	**Organizing Information**	Develops a Greedy algorithm to locally optimize information. Creates a certificate that demonstrates when a representation is optimal relative to the current information database.
Section 7	**Conclusion**	Concludes with comparisons to existing theories and possible applications, particularly in programming languages and broader academic knowledge management.

TABLE 2. Organization for the thesis.

2. Rationale

[TODO[MEDIUM]: make sure that the rationale blends nicely with some of these sections! For non-experts, say in logic, they should *not* need to read *how* the meta-theory works or such. They should only need to follow the bootstrap and understand the syntax.]

syntaaxIn this section, we justify the design of Welkin.

2.1. General Requirements.

The main purpose of an information base is to store information and answer user queries based on available information. To provide an upper bound on our requirements, we require two important properties:

1) The information base must be mechanized, which means only "effective operations" are allowed. We make "effective operations" precise with the technical notion of *partial computable functions*. Partial computable functions are computer programs that can process a finite input, and that either stop on an input or continue forever. The bedrock for these functions is the notion of a *Turing machine*, which models every current form of computation. For more information, see [Tur36]. To obtain universality, we need to express *every* one of these machines.
2) The information base must provide a mechanism to *ground* symbols defined by users. This stems from the Symbol Grounding Problem, posed by Harnard [Har90]. To avoid storing physical objects or such, users should have a way to provide *meaning* to symbols, or a way to *represent* their topic of interest.

Combined together, we can now state universality as follows: *an information base is universal if it can represent anything representable by a partial computable function.* We will clarify how we define "representable" in the next section.

2.2. Units.

[TODO[MEDIUM]: double check all examples!]

The first core concept in Welkin is the notion of a **handle**. Handles are left as free parameters in the theory[2] In an information base, handles are uniquely identified through their **key**, directly inspired by database theory. Keys are with triples of IDs $(\mathrm{UID}, \mathrm{RID}, \mathrm{HID})$, whose contents are the **user ID**, **revision ID**, and **handle ID**, respectively. Revisions ensure that storing axioms is **immutable**, or that a particular handle does not change over time. This is *not* a restriction on dynamic entities, but rather an indication to the information base to determine a particular version of a handle.

Example 2.2.0. In a scientific experiment, a handle could be an observation or experimental data. The unit is then written as a symbol, say u, and is *implicitly* bound to this meaning. To distinguish from other symbols, say v, the computational content is analyzed.[TODO[MEDIUM]: expand out this example!]

Indexing handles is only one important aspect of handles. Another is that handles are precisely by their relationships. These are written as **representations** $a - c \to b$, which means that a represents b in context c. Here, we call a the **sign**, b the **referent**,

[2]For mathematicians, this is analogous to how key notions are left undefined in Hilbert's formalism of geometry, or how a "set" is an undefined notion in set theory.

DRAFT

and c the **context**. One can interpret this characterization as follows: handles are defined by how they are and are *not* restricted. This idea is directly inspired from Fine's idea of arbitrary objects [Fin85], and allows refinements to a study of interest.

Example 2.2.1. A more looser example is a user written journal for therapy sessions, containing information about daily habits and emotions. While neither of these are stored in the information base, their handles are, via units habit and $\mathrm{emotions}$ in a context $\mathrm{journal}$. Moreover, multiple revisions of the journal can be made with dates or other unique IDs.

We represent handles in Welkin using an extended notion called a **unit**. A unit is either a handle, a finite collection of units, or a representation. Collections of units can have a uniquely generated name by the information base called a **block**[3].

Example 2.2.2.

Units define their own **context**, which means the unit provides its own names and representations. Writing $a - c \to b$, therefore, only applies to the *fixed* context c.

[TODO[SMALL]: use better names for entities/businesses/etc]

Example 2.2.3. A business could represent their operations using a unit $\mathrm{business}$ that contains units for their workers and ledgers. This allows another unit, say $\mathrm{business2}$, to contain its *own* label $\mathrm{workers}$ that is separate from the one in $\mathrm{business}$. In addition to separate labels, these contexts can have *distinct* rules, such as those for how business operations are performed.

2.3. Mechanizing Information.

From the notion of a unit, we practically define information on a unit v as a *unit that tracks reductions in* v. Formally, we define information as a proof used in derivations, which include refinements of v to other units. Key to this definition is making it *context-based*, thereby enabling separate truth-management systems (those systems defined by some Turing machine). This notion corresponds to Burgin's idea of information as an *operator* that transforms a system, and is closely to Bateson's famous quote that "information is a difference that makes a difference" [Bat00]. Our practical distinction between information and knowledge is that we *use* information, but users can assert their own notions of these terms by creating restricted contexts. There are technical constructions to show that our definition is *as general* as possible (Section 4), as well as optimally organized (Section 6).

There are two primary inference rules, which we informally describe now:

- **Internal Transitivity:** if u represents v in context c and v represents x in context c, then u represents x in context c.
- **Lifting:** if u represents v in context c and p represents q in context v, then within context c, p represents q in *nested* context u.

Transitivity is a common axiom in many systems. It is particularly efficient when used in contexts, as shown by several projects [Len+90], [GWW07]. Lifting is a notion closely tied to McCarthy's notion of lifting, but formulated with representations. The idea is, if a represents b in context c, then a provides an *abstraction* for b relative to c. Wherever b is involved in a relationship, a can be used *instead*, acting as a proxy relative

[3]In programming languages, this is analogous to an anonymous function.

DRAFT

to c. Contexts provide an import mechanism as well, thereby providing Welkin with a module based system. We provide a simplified version of one McCarthy's examples below. For more existing examples in the literature, refer to [BF95].

Example 2.3.4. (McCarthy's Above Theory). Consider a set of physical blocks. Denote the unit `above` to represent the relation `block A is above block B`. We want to say this could be `on` to mean `block A is on block B`, or `floating_above` to mean `block A is floating above block B`. One way to state is as two axioms: and we could state as an axiom: `A - on -> B` implies `A - above -> B` and `A - floating_above -> B` implies `A - floating_above -> B`. Lifting provides a more economical approach *without* needing to provide explicit terms, or provides a "point-free" rendering. We could that, within `above`, `above` represents `on` and `above` represents `floating_above`. In other words, `above` *precisely abstracts* from the precise relationship between the two blocks.

Another example concerns contexts in the presence of unique objects.

Example 2.3.5. Consider a biological survey of home pets, denoted by context `Survey`. Suppose there are two units in `Mammal`, say a dog `Fido` and a cat `Lucy`. We could state that `Fido` *represents* `Mammal`, to say that `Fido` acts as a stand-in for a mammal in the survey. The same could be said with `Lucy` representing `Mammal`. Note that, depending on the context, the referent need *not* be more refined than the sign; this is an intenional design choice for flexibility. With a another context `Taxonomy`, we may naturally state that `Mammal - Taxonomy -> Animal`, and we could say `Survey - instanceof -> Taxonomy`. Thus, the lifting rule implies in context `instanceof`, `Mammal - Survey -> Animal`. Using transitivity, we obtain `Fido - Survey -> Animal` and `Lucy - Survey -> Animal` in context `instanceof`. We can interpret this example as saying that the *relationships* of a general taxonomy are witnessed by a specific taxonomy, namely `Survey`, and that they propagate through contexts.[4]

The remaining rules are primarily for efficiency, enabling users to have positionally invariant information, see Definition 3.1.11.

2.4. **Base Operations.**

Now, units and information themselves could be expressed in infinitely many languages, with slightly different syntax or semantics. Welkin is carefully designed to be a *minimal* expression of these concepts, with minimal friction to express any other universal information base. These include:

- In ASCII, `a - b -> c` for representations. These can be interpreted *as* rewrite rules, depending on the context.
- Traditional braces `{ }` to denote closed definitions of contexts, inspired by the C programming language.
- Paths via dots `.` that is inspired by the Python programming language. Relative paths are denoted with multiple dots `...`, and absolute imports are prefixed with `#`. Subsets of units can be written via `u.{v, x}`, or even `u.{v --> x}` to refer to a subset of representations.

[4]In programming languages, particularly C++ [Str13] and Java [AGH05], this property is known as "upcasting", with a less refined base class being replaced with a more refined subclass. The reverse is "downcasting". Both are supported by Welkin for full expressivity.

- Imports are done through `@u`, which takes all subunits of `u` and puts them into the current scope. In other words, the implementation *implicitly* adds denotations `v <--> a.v` for each subunit `v` of `u`. This is motivated by the abundance of boolean logic in classical mathematics and computer science, as any finite circuit can be expressed in terms of `and` and `not`. Selecting specific subunits can be done via `@u.{v, x}`.
- Imports can be *negated* via the notation `~@u`, and specific units can be negated through `~u`. This is an uncommon feature of most programming languages, appearing primarily in Haskell and CSS. While potentially opaque, Welkin provides robust definitions to ensure that negated forms can be easily translated into more explicit ones *and* back again. [TODO[SMALL]: provide link, probably to bootstrap or so?]
- Comments *are* strings can be treated as any unit. No comments need to be removed in the files and can *enhance* the study of new subjects.[5] These units can *then* be analyzed by others to promote translations to other human languages, or be studied through the *overarching relationships* in the units.

In general, the minimal restricted keywords is crucial for providing support for other languages. An implementation of Welkin will contain a small section of ASCII encoding for easier standardization, but the rest of the program can be done *entirely* in the user's native language. This is a novel feature in most programming languages, which are either predominantly English or are fine tuned for specific human languages.

2.5. **Bootstrapping.**

There are two important considerations in Welkin: how the base operations are implemented, and how proofs are expressed. In previous revisions, the author separated these aspects, particularly when describing the syntax and semantics of the language. However, this left many proofs in English, making it difficult to transfer.

[TODO: provide citaitons] The author's emphasis of proofs stems from formal methods, the study of providing mathematical proof to verify computer programs. A central notion in this field is the **Trusted Computing Base (TCB)** of a program, which is code used to implement it, as well as the axioms used to prove its correctness. Many authors advocate for a small, fixed TCB. However, this is inefficient in practice… Setting for a single system, like ZFC, does not resolve whether Welkin is universal *for proofs*.

To resolve both issues, we "bootstrap" Welkin from the ground up. This will enable Welkin to define the syntax *as* a term in the language. We proceed as follows:

- We establish the base rules on units (Definition 3.1.12).
- Using English as our meta-language, we prove that our rules can define the S and K combinators as any combination of terms, and thus is Turing complete, (Theorem 3.1.1).
- We show the language has a natural unit that indexes every other index. This term will act as a verifier for proofs, which we will show does *not* limit which proofs we accept.

[5] Contexts can mimic regular comments, but Welkin aims to be inspectable by users. Encapsulation or private information can be enforced through *rejecting* specific contexts or only accepting certain ones.

DRAFT

1) Provide the base meta-theory for the language *in* Welkin. This will be PRA, written in a combinator based form. We will show the main axioms of PRA hold.
2) Prove that Peano Arithmetic is provably sound in a weak sense using a method developed by Artemov [Art24]. From there, we carry out Feferman's Reflection theorem written with our notions.
3) Define handles and information. We also justify information organization to promote the *most* optimal paths to prove things, *if* proofs are provided. This is deeply connected to Blum's speedup theorem and is essential for proving Goal 2.
4) Define the rules for path resolution and global IDs. These use a similar result to syntax descriptions and are significantly easier to read. [NOTE: emphasis on WIP here! Not yet developed in thesis !]
5) Finally, define the syntax through an *invertible syntax description*. This idea, pioneered by Rendel and Ostermann [RO10], provides a combinator approach to define how concrete syntax, that is user written, is converted into abstract syntax, an abstracted form, *and back*. Along with the syntax, the encoding is given *itself* through a representation, namely a table provided in this thesis.

3. Foundations

This section discuss the foundations of Welkin, as follows:

1) We define units and their rules.
2) We prove there is a unit that indexes every other unit (Section 3.2).
3) We define queries.

3.1. Base Rules.

As high-level notation, we write $a \equiv b$ to mean that a is definitionally equivalent to b. Note that this will not be defined in the syntax, see Section 2.5. Moreover, we will distinguish between *defining* a term as finite and *practically enforcing* it is finite. For a thorough discussion, see ?.

Definition 3.1.0. A **bit** is the symbols 0 or 1. A **binary word** is either the symbol ε (the **empty word**), and if w is a finite binary word, so are $w.0$ and $w.1$, where . is the symbol for **concatenation**. Nothing else is a word.

Definition 3.1.1. Equality $=$ and inequality $\neq$ on words w_1, w_2 is defined recursively:

- **Base case:** $\varepsilon = \varepsilon$.
- **Recursive step:** suppose $w_1 = b_1.w_{1'}$ and $w_2 = b_2.w_{2'}$, where b_1, b_2 as bits and $w_{1'}, w_{2'}$ are words. Then $w_1 = w_2$ if and only if $b_1 = b_2$ and $w_1' = w'2$. Moreover, $w_1 \neq w_2$ if and only if $b_1 \neq b_2$ or $w_1' \neq w_2'$.

Remark 3.1.2. Definition 3.1.8 is given constructively to ensure that if two finite words are unequal, then an explicit bit can act as a certificate for this inequality.

Definition 3.1.3. A *handle* is given by a **key**, a triple $(\mathrm{UID}, \mathrm{RID}, \mathrm{HID})$, where UID is a binary word called a **user ID**, RID is a binary word called the **revision ID**, and HID is a binary word called the **handle ID**. The interpretation of these parameters are free parameters, and therefore are outside the scope of this language.

DRAFT

Remark 3.1.4. Because handles act as free parameters, we work with them *through* truth. Based on this, the rest of the thesis will abstract away general units *as* sets of axioms. Free parameters are used to *define* Welkin itself, however, see Section 5.

[TODO[SMALL]: without getting into typing, enforce that equality has to be done on two words or two handles. Maybe lift to the latter to make sense?]

Definition 3.1.5. Consider two handles $h_1 \equiv (\mathrm{UID}_1, \mathrm{RID}_1, \mathrm{HID}_1), h_2 \equiv (\mathrm{UID}_2, \mathrm{RID}_2, \mathrm{HID}_2)$.

- h_1 and h_2 are equal, written $h_1 = h_2$, if and only if $\mathrm{UID}_1 = \mathrm{UID}_2$, $\mathrm{RID}_1 = \mathrm{RID}_2$, and $\mathrm{HID}_1 = \mathrm{HID}_2$.
- h_1 is not equal to h_2, written $h_1 \neq h_2$, if and only if at least one of the following hold: $\mathrm{UID}_1 \neq \mathrm{UID}_2$, $\mathrm{RID}_1 \neq \mathrm{RID}_2$, or $\mathrm{HID}_1 \neq \mathrm{HID}_2$.

To simplify the use of handle equality, we will ensure all defined handles are unique.

Definition 3.1.6. A **unit** is defined recursively as a finite combination of the following and nothing else:

- A handle, see Definition 3.1.9.
- A **block**, which is one of the following:
  - The symbol $\{\}$, the **empty block**.
  - Given a units g, u, $\{g, u\}$, $\{@g, u\}$, and $\{@g, \sim u\}$ are blocks, where $@g$ is a unit called the **expansion** of g and $\sim u$ is called the **exclusion** of u.
- A representation $a - c \to b$ of units a, b, c, where a is the **sign**, c is the **context**, and b is the **referent**. This is read as: a **represents** b **in context** c.

We will intentionally *avoid* defining equality on units and postpone this until Section 3.2.

[TODO[SMALL]: make sure to define this notation | recursively!]

For notation, we will set $a - c \to b \mid d$ to mean $\{a - c \to b, a - c \to d\}$, and $a \mid b - c \to d$ to mean $\{a - c \to d, b - c \to d\}$. This simplifies the presentation of the rules; we postpone introducing the operator $\mid$ into the syntax until Section 5. Moreover, we write $q \in c$ to mean $q - c \to c$. Notice that many-to-many relationships are allowed.

[TODO[SMALL]: provide labels/links.]

[TODO[SMALL]: Clarify role of global context!]

DRAFT

Definition 3.1.7. The following rules apply, stated over meta-variables a, b, c, d, g for units and h_1, h_2 for handles:

- **R1. Internal Transitivity:** $a - c \to b$ and $b - c \to d$ imply $a - c \to d$.
- **R2. Contextual Lifting:** $a - c \to b$ and $d - b \to g$ imply $\{d - a \to g\} \in c$.
- **R3. Empty:** $\{@g, \{\}\} \longleftrightarrow g$.
- **R4. Membership:** $\{@g, a\} \longleftrightarrow g$ if and only if $a - g \to a$.
- **R5. Identity:** $\{a\} \longleftrightarrow \{a - a \to a\} \longleftrightarrow a$.
- **R6. Expansion:** if $a - g \to b$, then $\{@g, c\} \leftrightarrow \{\{\{@g, a\}, b\}, c\}$.
- **R7. Exclusion:** if $g \longleftrightarrow \{@g, a\}$, then $\{\{@g, \sim a\}, b\} \longleftrightarrow \{@d, b\}$.
- **R8. Associativity:** $\{a, \{b, c\}\} \longleftrightarrow \{\{a, b\}, c\}$.
- **R9. Commutativity:** $\{\{a, b\}, c\} \longleftrightarrow \{\{a, c\}, b\}$.
- **R10. Null:** $\{a - \{\} \to b\} \longrightarrow \{\}$.
- **R11. Handle Substitution:** if handle h_1 is equal to handle h_2, then for any a, $\{a, h_1\} \longleftrightarrow \{a, h_2\}$.

We review the utility of each rule. Note that rules *between* contexts is entirely flexible and user defined. Moreover, only ? are needed for Turing completeness. We will show, however, that the remaining axioms are optimal when organizing information, see Section 6.

- **R1** and **R2** were discussed in Section 2.2.
- **R3** define the behavior of the empty unit $\{\}$, similar to the empty set.
- **R4** provides a way to represent $a \in g$ through the representation $a - g \to a$. Because of this, $a - g \to a$ may be a non-trivial path, so it is *not* required in every unit. Identity is expressed instead by $\{a - a \to a\}$, see the discussion below.
- **R5** reduces extraneous blocks. Note that this is *not* the same thing as the Quine atom, which states $\{a\} = a$ in a set theoretic context [Qui37]. We do *not* mean $a \in a$, but instead, $\{a\}$ is a *wrapper* around a. While not useful for handles, it is for specifying blocks of representations, such as $\{a - b \to c, b - c \to d\}$.
- **R6** and **R7** define how imports in the language work. An **import** is the process of joining the contents of one unit into another. **R6** says how an import can add new units in a block. **R7** provides a mechanism to *exclude* specified contents in g, which can themselves be representations. For example, $\{@\{a - b \to c, d\}, \sim\{a - b \to c\}\}$ reduces to $\{d\}$.
- **R8** and **R9** ensure that information can be repeated and is positionally invariant.
- **R10** provides a way to prevent representations in a block. We interpret this rule as stating: $\{\}$ contains nothing, so it cannot contain any representation, including $a - \{\} \to b$. We will use this rule to represent negation. Note that this is *distinct* from exclusion, as the latter is restricted to imports.
- **R11** enables equality in handles to pass through into representations. Besides this, note that equivalences on units are *entirely* user defined.

Before we proceed to prove Turing completeness, we introduce the SK-combinator calculus. This is an equational theory that is well known to be Turing complete [Cur30]. We provide a full definition as follows.

DRAFT

Definition 3.1.8. The SK-combinator calculus consists of the following:

- A **term** is defined recursively as either K or S, and if M, N are terms, so is (M) and their **application** MN.
- Evaluation: two terms M, N are **equal**, written $M = N$, if they can be deduced from the following axioms:
  - **Base Rules:** for all terms M, N, P:
    * $KMN = M$
    * $SMNP = MP(NP)$.
  - **Congruence:** if $M_1 = M_2$ and $M_2 = N = 2$, then $M_1 M_2 = N_1 N_2$.
  - **Equivalence:** $=$ forms an equivalence relation, which means for all M, N, P:
    * $M = M$.
    * $M = N$ if and only if $N = M$.
    * $M = N$ and $N = P$ imply $M = P$.

[TODO[SMALLL]: create boxes for remarks and important notes?]

Theorem 3.1.9. *Any partial computable function can be expressed as some unit.*

Proof. We prove that we can embed any term in the SK-combinator calculus, defined in Definition 3.1.13. This proof includes an important technique to represent *recursion*,

We express recursion via a unit L. For keys, we set:

- $L \equiv (1, 0, 0)$.
- $K \equiv (1, 0, 1)$, $S \equiv (1, 0, 2)$.
- $M \equiv (1, 0, 3)$, $N \equiv (1, 0, 4)$ $P \equiv (1, 0, 5)$.

For the rules, see Figure 1. Note that each rule of the form $A \longrightarrow B$ written in L means $A - L \to B$.[6]

$$
\begin{gathered}
L \equiv \{(K \longrightarrow L, S \longrightarrow L, \\
\{M \longrightarrow L\} \longrightarrow L, \\
\{N \longrightarrow L\} \longrightarrow L, \\
\{P \longrightarrow L\} \longrightarrow L, \\
C \longleftrightarrow \{N - M \to L\}, \\
C \longrightarrow \{M \longrightarrow L\}, \\
C \longrightarrow \{N \longrightarrow L\}, \\
\{Y - \{X - K \to L\} \to L\} - L \to X, \\
\{P - \{N - \{M - S \to L\} \to L\} \to L\} - L \to \{\{P - N \to L\} - \{P - M \to L\} \to L\})\}
\end{gathered}
$$

FIGURE 1. Definition of L [TODO: fix formatting!].

This definition means:

- L includes K and S as base cases. We interpret $K - L \to L$ to mean K is a term of L.

[6] As a note for logicians: these rules are extremely similar to a Hilbert-style proof system, with K and S corresponding to the rules $(\varphi \Rightarrow (\psi \Rightarrow \varphi))$ and $(\varphi \Rightarrow (\psi \Rightarrow \zeta)) \Rightarrow ((\varphi \Rightarrow \zeta) \Rightarrow (\psi \Rightarrow \zeta))$, respectively. This was one of Curry's insights in connecting logic to computation [Cur30].

DRAFT

- We include variables M, N, P over terms. The statement $\{M - L \to L\} - L \to L$ wraps around any term.
- Composition MN is represented as $N - M \to L$. Moreover, compositions MN can be broken down into its constituent parts M, N.
- The remaining representations are for the rule of K and S, respectively.

Now, L already includes the base rules for K and S. It remains to be shown that L is closed under composition: $M - L \to L$ and $N - L \to L$ imply $C - L \to L$, where $C \longleftrightarrow (N - M \to L)$. By using **R1** on $C - L \to (M - L \to L)$ and $(M - L \to L) - L \to L$, we obtain $C - L \to L$, completing the proof. □

3.2. **Base Recursor.**

The proof of Theorem 3.1.1 demonstrates how contexts enable powerful recursive definitions. However, the underlying construction is tedious and results in extremely verbose terms. Keys are assigned manually, which can easily be error prone. We will refine the proof with a **recursor** over units. This is a unit that indexes every unit, as well as every handle.

The recursor needs to index through all possible keys. We first need to define a unit to recurse over binary words; we provide a natural definition in Figure 2.

$$\begin{gathered}\text{bit} \longrightarrow 0 \mid 1 \\ \text{word} \longleftrightarrow \text{empty} \mid \{\text{top} \longrightarrow \text{bit}, \text{next} \longrightarrow \text{word}\}\end{gathered}$$

FIGURE 2. Generator for words in Welkin.

This is similar to the Lisp definition of a list. In detail:

- bit represents 0 or 1.
- word is recursively defined, as either empty or the pair $\{\text{top}, \text{next}\}$, where top is a bit and next is a word.

We need to include equality as well, refer to Definition 3.1.10:

[TODO[SMALL]]

$$\begin{gathered}\text{equality} \longleftrightarrow \{ \\ 0 \longleftrightarrow 0, \\ 1 \longleftrightarrow 1, \\ 0 \leftarrow \{\} \to 1, \\ \text{w1} \mid \text{w2} \longrightarrow \text{word}, \\ \{\text{w1} \longleftrightarrow \text{w2}\} \longleftrightarrow \{\text{w1} \longleftrightarrow \text{empty} \longleftrightarrow \text{w2}\} \\ \mid \{\}, \\ \text{h1} \mid \text{h2} \longrightarrow \text{handle}, \\ \}\end{gathered}$$

FIGURE 3. Definitions of equality in Welkin.

Lemma 3.2.10. *Figure 3 correctly implements equality*

Proof. TBD. □

From there, we can define handle IDs through triples, see Figure 4.

DRAFT

$$\text{handle} \longleftrightarrow \{\text{UID} \longrightarrow \text{word}, \text{RID} \longrightarrow \text{word}, \text{HID} \longrightarrow \text{word}\}$$

FIGURE 4. Generator for handle keys in Welkin.

Here, a handle is simply a pair $\{\text{UID}, \text{RID}, \text{HID}\}$, where UID, RID, HID are words.

Now, in Definition 3.1.12, we needed enough *separate* meta-variables. To do this in Welkin, we use representations of the form $\mathrm{u} \longrightarrow \mathrm{unit}$. This appeared frequently when defining terms in Theorem 3.1.1.

Definition 3.2.11. The **unit recursor** unit includes all rules in Definition 3.1.12, as well as the following in context unit:

- handle $\longrightarrow$ unit, see Figure 4.
- $\{\} \mid u \mid v \mid c \longrightarrow \text{unit}$.
- $\{u - c \to v\} \longrightarrow \text{unit}$.
- $\{u, v\} \mid \{@u, v\} \mid \{@u, \sim v\} \longrightarrow \text{unit}$.
- **Monotonicity:**
  - $u \longrightarrow \{@u, v\}$.
  - $\{@u, v\} - \{\} \to u$.
  - $\{@u, \sim v\} \longrightarrow u$.
  - $u - \{\} \to \{@u, \sim v\}$.

Remark 3.2.12. One interpretation of Definition 3.2.14 is defining the notion of *containment* between units, so unit could be written as $<$ in the language. Moreover, unit acts as the *maximum* of all units. Introducing monotoncitiy will be effective for optimizations, see Section 6.

The following statements are two parts of the same **Recursion theorem** for Welkin. The first two are straightforward; their proofs closely aligns with the definitions written in the meta-language (English).

Lemma 3.2.13. (Correctness). *For every unit u, $u - \text{unit} \to \text{unit}$.*

Proof. Fix the context to be unit. We proceed induction on units:

- **Base case:** this is immediate, as $\{\}$ and all handles are included.
- **Inductive step:** immediate; unit includes representations $a - b \to c$, as well as cases for $\{g, u\}$, $\{@g, u\}$, and $\{@g, \sim u\}$.

□

[TODO[SMALL]: rethink statement here.]

The last part behind the Recursion theorem underlies why unit is enough to embed the Welkin inside itself.

Lemma 3.2.14. (Uniqueness) *Suppose unit T contains exactly $u - T \to u$ for each unit u. Then $T \leftarrow R \to R$.*

Proof. Clearly $T - \text{unit} \to \text{unit}$, and $\text{unit} - \text{unit} \to T$ follows from observing that $\{\text{unit} - \text{unit} \to T\} \in T$. □

[TODO[MEDIUM]: clarify how unit is a verifier!]

An important consequence of the recursion theorem is a basic form of **reflection**. Most importantly, this theorem establishes that *any* way to characterize Welkin can be reduced to including the recursor unit. This establishes that unit is the *smallest* set of

DRAFT

foundations for Welkin. Note that unit acts as *both* a recursive procedure *and* verifier over units.

Theorem 3.2.15. (Base Reflection). *Let T be any unit extends* unit. *Then* $\{T \longrightarrow \mathrm{unit}\} - \mathrm{unit} \to \mathrm{unit}$.

Proof. We proceed by induction, fixing the base to be unit.

- **Base Case:** suppose T is a unit exactly with the rules $u - T \to T$ for every unit u and the rules in Definition 3.1.12. Then by Lemma 3.2.4, $T \longleftrightarrow \mathrm{unit}$, completing the base case.
- **Inductive step:** suppose $T = \{T', e\}$ for a units T', e where $T' \longrightarrow \mathrm{unit}$. Now, by monotonicitiy in unit, $T \longrightarrow T'$, hence by transitivity, $T \longrightarrow \mathrm{unit}$.

□

3.3. **Queries.**

By Theorem 3.1.1, every partial computable can be expressed as a unit. Additionally, in the construction used, reductions of terms are *also* represented. This provides a ceiling on what queries we can *express* computably. For more details, refer to [HMU07]. Roughly, let c be a context c.

Definition 3.3.16. Let c be a context. Then a **query over c** is a question over a fixed unit q: "Does $q - c \to q$ hold?"

To standardize this in Welkin, we will instead write:

$$\textit{Is } \{@c, \sim q\} \leftarrow \mathrm{unit} \to \{\}?$$

We can then apply unit in an attempt to evaluate this. In some cases, this is easy and only uses *direct definitions*. Consider, for example, examining p in context $\{p, q\}$. We can resolve this with a quick traversal, showing that p is in fact in $\{p, q\}$. However, Theorem 3.1.1 implies that finding general derivations are uncomputable.

We want information to *resolve* a query. Formally, we want information to be *proof*, not a "yes/no" answer by itself. When the context is finite, like $\{p, q\}$, there is a direct proof. But what about the general case? Some queries may reduce *forever*. While we cannot determine *which* ones reduce to $\{\}$ or not, we *can* find the ceiling of proofs expressible by computable functions. This process is more involved and is the subject of the next section.

4. Information

This section discusses the provably most general definition of information.

1) We overview Artemov's work in serial consistency. [Art24].
2) We establish the meta-theory, and show it can encompass any formal system.
3) We show the meta-theory "meta-proves" its own soundness, *without* contradicting known impossibility results.
4) We conclude with the definition of information.

Optimizations will be postponed to Section 6.

DRAFT

4.1. **Serial Soundness.**

A major goal in Welkin is to express any representable notion, *including* any representabale proof. Proofs are a finite certificate that consists of a valid sequence of steps. Importantly, proofs can be verified using a *total* computable function, one that can check *any* given proof. Because of the requirement for computable checking, limitations in computable functions apply. These are illustrated by several impossibility theorems in the literature, including Gödel's infamous incompleteness theorems [Göd06].

However, previous impossibility results rely on proving strong properties of proofs. Artemov argues that weaker, constructive properties can be used instead. We provide a high-level definition of **serial-consistency** before proceeding onto the embedding in Welkin.

Definition 4.1.0. Suppose T is sufficiently strong to express its own proofs. Then T is **serial-consistent** if there is a **selector** for T. A **selector** is a computable function s[7] with two properties:

- s accepts any string that encodes a valid proof.
- if s accepts a proof, then it cannot contain a contradiction.

Artemov overcomes Gödel's second incompleteness theorem. This theorem states that Peano Arithmetic PA, providing basic properties of the natural numbers, cannot prove its own consistency. Artemov instead demonstrates that PA proves its own *serial-consistency*, as follows:

- He uses a weak meta-theory, in this case, Primitive Recursive Arithmetic (PRA). This theory is bounded by a weak principle of induction.
- Artemov encodes all proofs as natural numbers in PRA. PRA is sufficiently strong *just* to encode all of these proofs.
- Finally, he uses a proof by induction, over encoded proofs in PRA, to show that PA proves each *individual* proof contains no contradictions. This is possible through partial truth definitions.

To be clear: Artemov does *not* show that there is a *single* proof that PA is consistent, that works for all proofs. Instead, his method *takes* a proof as a parameter. There have been extensive discussion on the validity of this technique, and its acceptance by other logicians.[8]

Despite its initial controversy, Artemov's techniques follow closely with several constructive schools, see [Art24]. This thesis builds upon this result with a stronger property: **serial-soundness**. The exact same proof applies, using the same partial truth definition. Tarski's theorem says that a formal system cannot define its own truth predicate *at the object language* [Tar83]. We circumvent this problem *precisely* with a meta-theory.

[TODO: explain the role of chains! It's different than reflection, so we should show *how* these chains can build up.]

[7]Technically, this is a primitive recursive function, but we will quickly generalize this to *any* computable function.

[8]The discussion is available online at: https://mathoverflow.net/questions/469247/situation-with-artemovs-paper.

DRAFT

Definition 4.1.1. The **meta recursor** meta over all units is defined recursively. We say that meta **meta-proves** T, denoted $T \mid -_{\mathrm{meta}}\, a$ if there is a proof through serial-soundness chains $\mathrm{unit}, T_1, ..., T_n$, where each unit T_i proves its own serial-soundness.

To show that this covers *every* proof that can be computably recognized, we need to discuss ordinals. We provide an embedding via unit.

Theorem 4.1.2. *The unit* meta *meta-proves that serial soundness implies soundness.*

Definition 4.1.3. The unit $\mathrm{recursive_ordinal}$ is defined recursively:

- **Base:**
- **Successor:**
- **Limit:**

Theorem 4.1.4. *The unit* meta *expresses any computably expressible proof. More precisely, for any recursive ordinal* α*, there is a unit* $T \in \mathrm{meta}$ *with greater theoretic proof ordinal than* α

Proof.

□

Note, however, that detecting if an ordinal is recursive is undecidable. The upper bound is the best one can hope for, in general.

As an immediate corollary, this proves that unit, being a simple verifier, does *not* impose further restrictions on proofs.

Theorem 4.1.5. *The unit* meta *meta-proves its own serial soundness, and hence, soundness.*

4.2. **Defining Information.**

This section provides the definition of information in Welkin.

Definition 4.2.6. Information about a query q in context c is a any partial meta-proof in meta of a derivation of $\{@c, q\}$.

5. Syntax

We keep this section self-contained with explicit alphabets and recursive definitions. For consistency with Welkin, we write syntax using `type-writer font`. Notationally, we write $a_0, ..., a_n$ for a finite list of items, and use $a \coloneqq a_1 \mid ... \mid a_n$ to denote a definition of a in terms of $a_1, ..., a_n$. For verification purposes, we will incorporate fixed bounds and completely unambiguous notation into Section 3.

5.1. **Words.**

Welkin's main encoding uses binary words, but add notation for decimal and hexadecimal.

```
bit ::= 0 | 1
digit ::= 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
nibble ::= A | B | C | D | E | F
```

Listing 1. Binary, decimal, and hexadecimal digits.

DRAFT

A word is a sequence of digits, see Listing 2. We leave concatenation `.` as an undefined notion. We set concatenation to be right-associative, i.e., `(w.w').w'' = w.(w'.w'')`, and abbreviate `w.w'` as `w w'`.

```
word ::= binary | decimal | hex
binary ::= bit | binary.bit
decimal ::= digit | decimal.digit
hex ::= nibble | hex.nibble
```

LISTING 2. Definition of words.

Equality is defined recursively, shown in Listing 3. Note that leading zeros are allowed.

[TODO[MEDIUM]: determine how to make Listing 3 work well in text **and** Welkin.]

```
0.word = word
0b0.word = word
0x0.word = word
```

LISTING 3. Definition of word equality.

5.2. **Terminals.**

[TODO[SHORT]: determine whether to put these details in appendix. Maybe given the *overarching* grammar, at a high level?] Welkin uses ASCII as its base encoding. The term ASCII is slightly ambiguous, as there are subtly distinct variants, so we formally define US-ASCII as a standard version.[9]

Definition 5.2.0. US-ASCII consists of 256 symbols, listed in Table 3.

To represent general encodings, there is a binary format supported for strings, see Listing 5.

DRAFT

[9]Note that this table *itself* is a representation, which represents glyphs with binary words. The use of these kinds of representations occur frequently in Welkin, see Section 3.

Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph	Dec.	Hex.	Glyph
0	00	NUL	32	20	Space	64	40	@	96	60	`
1	01	SOH	33	21	!	65	41	A	97	61	a
2	02	STX	34	22	"	66	42	B	98	62	b
3	03	ETX	35	23	#	67	43	C	99	63	c
4	04	EOT	36	24	$	68	44	D	100	64	d
5	05	ENQ	37	25	%	69	45	E	101	65	e
6	06	ACK	38	26	&	70	46	F	102	66	f
7	07	BEL	39	27	'	71	47	G	103	67	g
8	08	BS	40	28	(	72	48	H	104	68	h
9	09	HT	41	29	)	73	49	I	105	69	i
10	0A	LF	42	2A	*	74	4A	J	106	6A	j
11	0B	VT	43	2B	+	75	4B	K	107	6B	k
12	0C	FF	44	2C	,	76	4C	L	108	6C	l
13	0D	CR	45	2D	-	77	4D	M	109	6D	m
14	0E	SO	46	2E	.	78	4E	N	110	6E	n
15	0F	SI	47	2F	/	79	4F	O	111	6F	o
16	10	DLE	48	30	0	80	50	P	112	70	p
17	11	DC1	49	31	1	81	51	Q	113	71	q
18	12	DC2	50	32	2	82	52	R	114	72	r
19	13	DC3	51	33	3	83	53	S	115	73	s
20	14	DC4	52	34	4	84	54	T	116	74	t
21	15	NAK	53	35	5	85	55	U	117	75	u
22	16	SYN	54	36	6	86	56	V	118	76	v
23	17	ETB	55	37	7	87	57	W	119	77	w
24	18	CAN	56	38	8	88	58	X	120	78	x
25	19	EM	57	39	9	89	59	Y	121	79	y
26	1A	SUB	58	3A	:	90	5A	Z	122	7A	z
27	1B	ESC	59	3B	;	91	5B	[	123	7B	{
28	1C	FS	60	3C	<	92	5C	\	124	7C	|
29	1D	GS	61	3D	=	93	5D	]	125	7D	}
30	1E	RS	62	3E	>	94	5E	^	126	7E	~
31	1F	US	63	3F	?	95	5F	_	127	7F	DEL

DRAFT

TABLE 3. US-ASCII codes and glyphs.

We denote specific characters through quotes, escaping if necessary. There are several important character classes in Listing 4, denoted through double quotes.

```
PRINTABLE  ::= [0x20-0x7E]
WHITESPACE ::= [0x09, 0x0A, 0x0D, 0x20]
DELIMITER ::= [0x7B, 0x7D, 0x2C, 0x2D, 0x2A, 0x3C, 0x3E, 0x22, 0x27,
0x5C, 0x7D]
```

LISTING 4. Important character classes.

Strings allow escaped single or double quotes, see Listing 5. IDs are special cases of strings that do not require quotes but forbid whitespace and certain characters, see Listing 6.

```
STRING ::= SQ_STRING | DQ_STRING
SQ_STRING ::= "'" (SQ_CHAR | ESCAPE_SQ )* "'"
DQ_STRING ::= "'" (DQ_CHAR | ESCAPE_DQ )* "'"

SQ_CHAR ::= PRINTABLE \ {'}
DQ_CHAR ::= PRINTABLE \ {"}
ESCAPE_SQ ::= "\'" | "\\"
ESCAPE_DQ ::= "\"" | "\\"
```

LISTING 5. Strings.

```
IMPORT ::= "@" ID
ID :: ID_CHAR+
ID_CHAR ::= PRINTABLE / (DELIMITERS + WHITESPACE + "#" + "@" + "'" +
"\"")
```

LISTING 6. IDs.

[TODO(MEDIUM): find good way to implement directly with Welkin or discuss.]

5.3. **EBNF Notation and Parse Trees.**

We define our variant of EBNF below:

Definition 5.3.1. An **EBNF** grammar consists of **productions**, which are pairs of the form `r ::= a_1 ... a_n`. On the right-hand side, juxtaposition means concatenation.

- Uppercase names require *no* whitespace between them. Otherwise, whitespace is allowed.
- `a ::= a_1 | ... | a_n` is short-hand for `{a ::= a_i | 1 <= i <= n}`.
- `(a_1)*` means zero or more instances of a_1.
- `(a_1)+` means one or more instances of a_1.
- `(a_1)?` means zero or one instance of a_1.

5.4. **The Welkin Grammar.**

Welkin's grammar is displayed in Listing 7, inspired by a minimal, C-style syntax. Note that when concatenating two terminals, denoted in uppercase, no whitespace between them is allowed, but in any other case, any amount of whitespace is allowed but ignored.

DRAFT

```
start        ::= terms
terms        ::= term ("," term)* ","? | EPS
term         ::= arc | graph | tuple | path
arc          ::= (term ("-" | "<-") term ("-" | "->"))+ term
graph        ::= path? "{" terms "}"
tuple        ::= path? "(" terms ")"

path         ::= MODIFIER? path_segment* unit
path_segment ::= unit | ".*" | "."+
unit         ::= ID | STRING


MODIFIER ::= "#" |  "@" | "~@" | "~"
ID         ::= ID_CHAR+
ID_CHAR    ::= PRINTABLE \ (DELIMITERS | WHITESPACE | "#" | "@" | "~" |
"'" | '"')
DELIMITERS ::= "," | "." | "-" | "<" | ">" | "*" | "(" | ")" | "{" | "}"
STRING     ::= SQ_STRING | DQ_STRING
SQ_STRING  ::= "'" (SQ_CHAR | ESCAPE_SQ )* "'"
DQ_STRING  ::= '"' (DQ_CHAR | ESCAPE_DQ )* '"'
SQ_CHAR    ::= PRINTABLE \ {"'"}
DQ_CHAR    ::= PRINTABLE \ {'"'}
ESCAPE_SQ  ::= "\'" | "\\"
ESCAPE_DQ  ::= '\"' | "\\"
PRINTABLE  ::= [0x20-0x7E]
WHITESPACE ::= [0x09, 0x0A, 0x0D, 0x20]
DELIMITER ::= [0x7B, 0x7D, 0x2C, 0x2D, 0x2A, 0x3C, 0x3E, 0x22, 0x27,
0x5C, 0x7D]
EPS        ::= ""
```

LISTING 7. The grammar for Welkin. The terminals `id` and `string` are defined in Listing 2 and Listing 5, respectively

[TODO(HIGH): determine if this is good as is or needs to be put into bootstrap!]

5.5. **Proof of Unambiguity.**

We now prove that the Welkin language is unambiguous by showing it is LL(1), a rich class of grammars that can be efficiently parsed. For more details, please consult [Aho+06].

Moreover, we define the top of a word in Listing 8.

```
top(word) ::= nil => nil | bit.word => bit
```

LISTING 8. Definition of the top of a word.

Definition 5.5.2. ([RS70]). A grammar is LL(1) iff the following holds: for any terminal w_1 and nonterminal A, there is at most one rule r such that for some w_2, w_3 appearing at the top of A such that,

- $S \Rightarrow \mathrm{top}(w_1)Aw_3$
- $A \Rightarrow w_2(p)$
- $\mathrm{top}(w_2w_3) = w$

Theorem 5.5.3. *There exists some* LL(1) *grammar that accepts the same strings as the Welkin grammar Listing 7. Hence, Welkin's syntax is unambiguous, i.e., every string accepted by the language has exactly one derivation.*

DRAFT

Proof. We use transformations in Table 4 that preserve the language of the original grammar, resulting in Listing 9. For the refactor step by step, see Table 5. We can readily verify that there are no shared prefixes for a single production, see Table 6. Because there are no conflicts, the transformed grammar is LL(1), and hence, the grammar is unambiguous.

Rule ID	Name	Description
T0	**Group Flattening**	Converts Kleene stars `A*` and regex-like lists into right-recursive forms `A' ::= A A' | EPS`.
T1	**Left Refactoring**	Transforms overlapping prefixes `A ::= B C | B D` into `A ::= B (C | D)` to eliminate FIRST set collisions.
T2	**Lexical State Expansion**	Expands complex sequence operators (`+`, `*`) into strict right-recursive terminal rules, ensuring contiguous consumption without whitespace interruptions.
T3	**Left-Recursion Removal**	Eliminates immediate left-recursion `A ::= A B | C` by rewriting as `A ::= C A'` and `A' ::= B A' | EPS` to prevent infinite loops.

TABLE 4. Well known transformations on grammars that preserve string acceptance.

DRAFT

```
start      ::= terms
terms      ::= term terms_tail | EPS
terms_tail ::= "," terms | EPS
term       ::= node chain

chain      ::= left_link node right_link
               node chain | EPS

left_link  ::= "-" | "<-"
right_link ::= "-" | "->"

path     ::= path_segment* unit
path_segment ::= MODIFIER? (UNIT | ".*" | "."+)

node       ::= PATH opt_block | block
opt_block  ::= block | EPS
block      ::= "{" terms "}"
             | "(" terms ")"

PATH       ::= MODIFIER PATH_BODY
             | PATH_BODY
PATH_BODY  ::= "." PATH_DOTS
             | UNIT PATH_TAIL
PATH_DOTS  ::= "*" PATH_BODY
             | "." PATH_DOTS
             | UNIT PATH_TAIL
PATH_TAIL  ::= PATH_BODY | EPS

UNIT     ::= IMPORT | ID | STRING
MODIFIER   ::= "#" | "@" | "~@" | "~"
ID         ::= ID_CHAR+
ID_CHAR    ::= PRINTABLE \ (DELIMITERS | WHITESPACE | "#" | "@" | "~" |
"'" | '"')
DELIMITERS ::= "," | "." | "-" | "<" | ">" | "*" | "(" | ")" | "{" | "}"
STRING     ::= SQ_STRING | DQ_STRING
SQ_STRING  ::= "'" (SQ_CHAR | ESCAPE_SQ )* "'"
DQ_STRING  ::= '"' (DQ_CHAR | ESCAPE_DQ )* '"'
SQ_CHAR    ::= PRINTABLE \ {"'"}
DQ_CHAR    ::= PRINTABLE \ {'"'}
ESCAPE_SQ  ::= "\'" | "\\"
ESCAPE_DQ  ::= '\"' | "\\"
EPS        ::= ""
```

LISTING 9. Transformed LL(1) grammar for Welkin, with all terminals defined.

DRAFT

Original	Transform	LL(1)

`start ::= terms,`
`terms ::= term (","  term)*`
`","? | EPS`
	Transform 1	`start ::= terms`
		`terms ::= term`
		`terms_tail | EPS`
		`terms_tail ::= "," terms |`
		`EPS`

`term ::= arc | graph |`
`group | path`
`arc ::= (term ("-" | "<-")`
`term ("-" | "->"))+`
`term`
	Transform 4	`/* Extracted 'node' to fix`
		`recursion.`
		`Arcs are strict left/right`
		`link pairs */`
		`term ::= node chain`
		
		`chain ::= left_link node`
		`right_link`
		`node chain |`
		`EPS`
		
		`left_link ::= "-" | "<-"`
		`right_link ::= "-" | "->"`

`graph ::= path? "{" terms "}"`
`tuple ::= path? "(" terms ")"`
`path ::= modifier?`
`path_segment* unit`
	Transform 2,
	Transform 3
		`/* Left-factor path & blocks.`
		`*/`
		`node ::= PATH opt_block`
		`| block`
		`opt_block ::= block | EPS`
		`block ::= "{" terms "}"`
		`| "(" terms ")"`
		
		`/* Expand path +, *`
		`contiguously */`
		`PATH ::= MODIFIER`
		`PATH_BODY`
		`| PATH_BODY`
		`PATH_BODY ::= "." PATH_DOTS`
		`| UNIT PATH_TAIL`
		`PATH_DOTS ::= "*" PATH_BODY`
		`| "." PATH_DOTS`
		`| UNIT PATH_TAIL`
		`PATH_TAIL ::= PATH_BODY |`
		`EPS`

TABLE 5. Refactor of grammar Listing 7 into Listing 9. Entries with - mean that no changes are needed.

DRAFT

Non-Terminal	Lookahead (a)	Production Chosen
start	"#" | "@" | "~@" | "~" | "." | ID | STRING | "{" | "(" | EOF	terms
terms	"#" | "@" | "~@" | "~" | "." | ID | STRING | "{" | "("	term terms_tail
	EOF | "}" | ")"	EPS
terms_tail	","	"," terms
	EOF | "}" | ")"	EPS
term	"#" | "@" | "~@" | "~" | "." | ID | STRING | "{" | "("	node chain
node	"#" | "@" | "~@" | "~" | "." | ID | STRING	PATH opt_block
	"{" | "("	block
opt_block	"{" | "("	block
	EOF | "}" | ")" | "," | "-" | "<-" | "->"	EPS
block	"{"	"{" terms "}"
	"("	"(" terms ")"
chain	"-" | "<-"	left_link node right_link node chain
	EOF | "}" | ")" | ","	EPS
left_link	"-"	"-"
	"<-"	"<-"
right_link	"-"	"-"
	"->"	"->"
PATH	"#" | "@" | "~@" | "&"	MODIFIER PATH_BODY
	"." | ID | STRING	PATH_BODY
PATH_BODY	"."	"." PATH_DOTS
	ID | STRING	UNIT PATH_TAIL
PATH_DOTS	"*"	"*" PATH_BODY
	"."	"." PATH_DOTS
	ID | STRING	UNIT PATH_TAIL
PATH_TAIL	"." | ID | STRING	PATH_BODY
	EOF | "}" | ")" | "," | "-" | "<-" | "->" | "{" | "("	EPS
MODIFIER	"#"	"#"
	"@"	"@"
	"~@"	"~@"
	"&"	"&"
UNIT	ID	ID
	STRING	STRING

TABLE 6. LL(1) Table for Listing 9

□

6. ORGANIZING INFORMATION

This section overview the most optimal organization of Welkin. We proceed in several parts:

1) We define metrics on units, and prove that *any* metric can be converted into one on size.

DRAFT

2)

6.1. Computable Measures and Size.

[TODO[MEDIUM]: incorporate numbers from syntax to make this easier!]

We adapt Blum's axioms [Blu67] for measures on units.

Definition 6.1.0. A **computable measure** is a provably computable function Φ that assigns units to natural numbers such that bounds on Φ are computable on *any* input.

[TODO[SMALL]: define size for units based ona simple representation. We can ensure this is *small* to have small overhead.]

Definition 6.1.1. The **size** of a unit is defined recursively:

Theorem 6.1.2. *Any metric satisfying Blum's axioms can be converted into one about size (the number of bytes in a unit).*

Remark 6.1.3. Of course, the conversion *may* be extremely large, so this *too* can be queried to find a more optimal approach. In any case, finding a constant time apparatus to work with these conversions is more praactical and can be automated by Welkin.

6.2. Localized Size Compression.

Welkin focuses on optimizing size as that is the simplest metric to use. The algorithm to calculate size is straightforward, based on the *direct* representations of units. Thus, one can provide a certificate about whether information about a context is *more compressed* than another.

The foundational unit recursor is meant to be a simple, direct verifier of proofs. However, it is *not* efficient for every problem. Going back to Theorem 3.1.1, there can easily be large expansions from the SK combinators. How do we tailor recursors *for* a specific context?

6.3. Query Compression.

Combining the previous sections, we define how to query on compressed representations, generalizing ?.

7. Conclusion

This thesis introduced Welkin, a universal, formalized information language. The syntax (Section 5) was defined rigorously with a small EBNF, shown to be accepted by an LL(1) grammar, showing that parsing is unambiguous. The semantics (Section 3) were provided with several passes to convert parse trees into units, which contain both a hierarchical and relational structure for scoping and direct representations, respectively. Units have key properties that enable them to express any partial computable function Theorem 3.1.1, in conjungtion with expressing any truth management system, demonstrates **universality** of the system. This is practically demonstrated by showing that all the major paradigms in Information Management and Knowledge Management are expressible within Welkin. Moreover, it was shown that there is a way to best organize the language given available information Section 6, showing **scalability**. Finally, the bootstrap in Section 3 self-hosts the language within a bounded 64 variant, whose complete Unambiguity (as well as the grammar's prior) establishes **standardization**. Revisions further enhance this by

DRAFT

The remaining sections show several areas for future work. This list is not exhaustive and, by the previous arguments, and can be applied to *any* subject with computable representations (essentially, any human subject).

7.1. **Comparisons to Existing Literature.**

In addition to domain specific proposals, there are approaches for general IM which still fail to resolve all issues. One prominent example is Burgin's General Theory of Information [Bur09] that comprehensively includes many separate areas for IM, including the complexity-based Algorithmic Information Theory, through a free parameter called an "infological system", which encompasses domain specific terminology and concepts. In contrast to other approaches, Burgin's generalized theory is flexible and enables greater coverage of different kinds of information [Mik23]. Despite this coverage Burgin does not closely tie the free parameter with his formal analysis of Algorithmic Information Theory, making it unclear how to use this in a practical implementation. Burgin's Theory of Information, along with broad proposals, have severe shortcomings, highlighting major obstacles for IM.

7.2. **Programming Languages and Formal Verification.**

[TODO: add discussions on supporting interoperability with languages, providing more robust and unified implementations of core libraries, compilers, and operating systems.]

Taking the enable custom hardware implementations for checking and reduce the surface area for attacks on verifying certifications for many applications, including cryptography.

Moreover, the proposed architecture could use an LLM as an oracle.

7.3. **Mathematical and Scientific Knowledge.**

[TODO: discuss more applications of Welkin in depth.]

There are several possible projects to pursue in mathematics and scientific research. For mathematics, there are several existing rojects for storing mathematical information (see [CF09] for more details). Older proposals, including the QED Manifesto [KR16] and the Module system for Mathematical Theories (MMT), aimed to be more general and have seen limited success. More centralized systems, like `mathlib` in the Lean proof assistant [The20], have seen adoption but do not give equal coverage nor are interoperable with other systems. Welkin enables this interopreeable through gradual translations, and with Section 6, one can always determine if there is enough *direct* information to complete a translation. This will help facilitate reusability among major tools, and aid in formal verification (Section 7.2) well.

Along with mathematics, Welkin could provide more rigorous frameworks for the sciences, with are currently scattered with different proposals. One prominent proposal is the Findable Accessible Interoperable Reusable (FAIR) guidelines [Wil+16]. Instead of providing a concrete specification or implementation, FAIR provides best practices for storing scientific information. However, multiple papers have outlined problems with these overarching principles, including missing checks on data quality [Gui+25], missing expressiveness for ethics frameworks [Car+21], and severe ambiguities that affect implementations [Jac+20]. Welkin addresses these by using contexts strategically. Experiments can be compared using revisions, and disagreements

DRAFT

between experts can be analyzed using separate contexts. These contexts can then *distinguish* between different theories, and scientists can select the unit with the best or most comprehensive evidence. Metrics for such evidence can be *representable* to a certain point, but at a minimum, they can be more effectively analyzed.

7.4. **Humanities.**

[TODO: significantly expand.]

Information Management in the humanities has few models, including an adaption of FAIR [Har+20] and discipline specific, linked databases in the PARNTHEOS project [Hed+19]. Welkin could assist by providing a space to help standardize this and localize different publication styles and literary theories.

References

[Ack89] Ackoff, R L, From Data to Wisdom, Journal of Applied Systems Analysis **16** (1989) 3–9.

[Aho+06] Aho, Alfred V., Lam, Monica S., Sethi, Ravi, and Ullman, Jeffrey D., *Compilers: Principles, Techniques, and Tools (2nd Edition)*, Addison Wesley, 2006.

[Ang18] Angles, Renzo, The Property Graph Database Model., in *AMW* (Olteanu, Dan and Poblete, Barbara (eds.), CEUR-WS.org.

[AGH05] Arnold, Ken, Gosling, James, and Holmes, David, *The Java Programming Language*, 4th ed., The Java Series, Addison-Wesley Professional, Upper Saddle River, NJ, 2005.

[Art24] Artemov, Sergei, Serial properties, selector proofs and the provability of consistency, Journal of Logic and Computation **35** no. 3 (2024) exae34.

[Atk23] Atkin, Albert, Peirce's Theory of Signs, Spring2023 ed., Metaphysics Research Lab, Stanford University, 2023.

[Bat00] Bateson, Gregory, *Steps to an ecology of mind*, University of Chicago Press ed., University of Chicago Press, Chicago, 2000.

[Blu67] Blum, Manuel, A Machine-Independent Theory of the Complexity of Recursive Functions, J. Acm **14** no. 2 (1967) 322–336.

[Buc91] Buckland, Michael K., Information as thing, Journal of the American Society for Information Science **42** no. 5 (1991) 351–360.

[Bur09] Burgin, Mark, *Theory of Information*, World Scientific, 2009.

[BF95] Buvač, Saša, and Fikes, Richard, A declarative formalization of knowledge translation, in *Proceedings of the Fourth International Conference on Information and Knowledge Management*, Association for Computing Machinery, Baltimore, Maryland, USA, pp. 340–347.

[CF09] Carette, Jacques, and Farmer, William M., A Review of Mathematical Knowledge Management, in *Intelligent Computer Mathematics* (Carette, Jacques, Dixon, Lucas, Coen, Claudio Sacerdoti, and Watt, Stephen M. (eds.), Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 233–246.

[Car+21] Carroll, Stephanie Russo, Herczog, Edit, Hudson, Maui, Russell, Keith, and Stall, Shelley, Operationalizing the CARE and FAIR Principles for Indigenous Data Futures, Scientific Data **8** no. 1 (2021) 108.

[Cur30] Curry, H. B., Grundlagen der Kombinatorischen Logik, American Journal of Mathematics **52** no. 3 (1930) 509–536.

[Dic25] Dictionary, Oxford English, welkin, n., Oxford University Press, 2025.

[Fin85] Fine, Kit, *Reasoning with Arbitrary Objects*, Blackwell, New York, NY, USA, 1985.

[Flo10] Floridi, Luciano, *Information : a very short introduction*, Oxford University Press, Oxford ; New York, 2010.

[GWW07] Goczyla, Krzysztof, Waloszek, Wojciech, and Waloszek, Aleksander, Contextualization of a DL Knowledge Base., p. .

[Gui+25] Guillen-Aguinaga, Miriam, Aguinaga-Ontoso, Enrique, Guillen-Aguinaga, Laura, Guillen-Grima, Francisco, and Aguinaga-Ontoso, Ines, Data Quality in the Age of AI: A Review of Governance, Ethics, and the FAIR Principles, Data **10** no. 12 (2025).

DRAFT

[Göd06] Gödel, Kurt, Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I, Monatshefte für Mathematik **149** no. 1 (2006) 1–29.

[Har90] Harnad, Stevan, The Symbol Grounding Problem, Physica D **42** (1990) 335–346.

[Har07] Harrington, Brian, ASKNet: Automated Semantic Knowledge Network., in *AAAI*, AAAI Press, pp. 1862–1863.

[Har+20] Harrower, Natalie, Quinn, Mary, Tóth-Czifra, Erzsébet, and others, *Sustainable and FAIR Data Sharing in the Humanities: Recommendations of the ALLEA E-Humanities Working Group*, Berlin, 2020.

[HS13] Hazarika, Shyamanta, and Sinha, Smriti, Formal concept analysis: current trends and directions, Artificial Intelligence Review **44** (2013) 1–40.

[Hed+19] Hedges, Mark, Stuart, David, Tzedopoulos, George, Bassett, Sheena, Garnett, Vicky, Giacomi, Roberta, and Sanesi, Maurizio, *Digital Humanities Foresight: The future impact of digital methods, technologies and infrastructures*, GOEDOC, Dokumenten-und Publikationsserver der Georg-August-Universität Göttingen, 2019.

[Hit21] Hitzler, Pascal, A review of the semantic web field, Commun. ACM **64** no. 2 (2021) 76–83.

[Hit+09] Hitzler, Pascal, Krötzsch, Markus, Parsia, Bijan, Patel-Schneider, Peter F., and Rudolph, Sebastian, *OWL 2 Web Ontology Language Primer*, 2009.

[HMU07] Hopcroft, John E., Motwani, Rajeev, and Ullman, Jeffrey D., *Introduction to Automata Theory, Languages, and Computation*, Third, Pearson/Addison Wesley, Boston, 2007.

[Jac+20] Jacobsen, Annika, Miranda Azevedo, Ricardo de, Juty, Nick, Batista, Dominique, Coles, Simon, Cornet, Ronald, Courtot, Mélanie, Crosas, Mercè, Dumontier, Michel, Evelo, Chris T., Goble, Carole, Guizzardi, Giancarlo, Hansen, Karsten Kryger, Hasnain, Ali, Hettne, Kristina, Heringa, Jaap, Hooft, Rob W.W., Imming, Melanie, Jeffery, Keith G., Kaliyaperumal, Rajaram, Kersloot, Martijn G., Kirkpatrick, Christine R., Kuhn, Tobias, Labastida, Ignasi, Magagna, Barbara, McQuilton, Peter, Meyers, Natalie, Montesanti, Annalisa, Reisen, Mirjam van, Rocca-Serra, Philippe, Pergl, Robert, Sansone, Susanna-Assunta, Silva Santos, Luiz Olavo Bonino da, Schneider, Juliane, Strawn, George, Thompson, Mark, Waagmeester, Andra, Weigel, Tobias, Wilkinson, Mark D., Willighagen, Egon L., Wittenburg, Peter, Roos, Marco, Mons, Barend, and Schultes, Erik, FAIR Principles: Interpretations and Implementation Considerations, Data Intelligence **2** nos. 1–2 (2020) 10–29.

[KBp] KBpedia, OpenCyc Resources.

[KR16] Kohlhase, Michael, and Rabe, Florian, QED Reloaded: Towards a Pluralistic Formal Library of Mathematical Knowledge, Journal of Formalized Reasoning **9** (2016) 201–234.

[Len95] Lenat, Douglas B, CYC: A large-scale investment in knowledge infrastructure, Communications of the ACM **38** no. 11 (1995) 33–38.

[Len+90] Lenat, Douglas B., Guha, R. V., Pittman, Karen, Pratt, Dexter, and Shepherd, Mary, Cyc: toward programs with common sense, Commun. ACM **33** no. 8 (1990) 30–49.

[The20] The mathlib Community, The lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, Association for Computing Machinery, New Orleans, LA, USA, pp. 367–381.

[McC93] McCarthy, John, Notes on formalizing context, in *Proceedings of the 13th International Joint Conference on Artifical Intelligence - Volume 1*, Morgan Kaufmann Publishers Inc., Chambery, France, pp. 555–560.

[Mes+15] Mesgari, Mostafa, Okoli, Chitu, Mehdi, Mohamad, Nielsen, Finn Årup, and Lanamäki, Arto, ``The sum of all human knowledge'': A systematic review of scholarly research on the content of Wikipedia, Journal of the Association for Information Science and Technology **66** no. 2 (2015) 219–245.

[Mik23] Mikkilineni, Rao, Mark Burgin's Legacy: The General Theory of Information, the Digital Genome, and the Future of Machine Intelligence, Philosophies **8** no. 6 (2023).

[MDV26] Mora, Angel, Dominguez, Jesus, and Villalon, Ana, fcaR: Formal Concept Analysis, 2026.

[Qui37] Quine, W. V., New Foundations for Mathematical Logic, The American Mathematical Monthly **44** no. 2 (1937) 70–80.

[Rei17] Reisman, Miriam, EHRs: The Challenge of Making Electronic Data Usable and Interoperable, P & T : a peer-reviewed journal for formulary management **42** (2017) 572–575.

[RO10] Rendel, Tillmann, and Ostermann, Klaus, Invertible syntax descriptions: unifying parsing and pretty printing, in *Proceedings of the Third ACM Haskell Symposium on Haskell*, Association for Computing Machinery, Baltimore, Maryland, USA, pp. 1–12.

[RWE15] Robinson, Ian, Webber, Jim, and Eifrem, Emil, *Graph Databases*, 2nd ed., O'Reilly Media, Inc., 2015.

[RS70] Rosenkrantz, D.J., and Stearns, R.E., Properties of deterministic top-down grammars, Information and Control **17** no. 3 (1970) 226–256.

[SL17] Smith, Brent, and Linden, Greg, Two Decades of Recommender Systems at Amazon.com, IEEE Internet Computing **21** no. 3 (2017) 12–18.

[Str13] Stroustrup, Bjarne, *The C++ Programming Language*, 4th ed., Addison-Wesley Professional, Boston, MA, 2013.

[Tar83] Tarski, Alfred, The Concept of Truth in Formalized Languages, Hackett Publishing Co., Indianapolis, IN, 1983.

[Tur36] Turing, Alan M., On Computable Numbers, with an Application to the Entscheidungsproblem, Proceedings of the London Mathematical Society **2** no. 42 (1936) 230–265.

[Wil+16] Wilkinson, Mark D., Dumontier, Michel, Aalbersberg, IJsbrand Jan, Appleton, Gabrielle, Axton, Myles, Baak, Arie, Blomberg, Niklas, Boiten, Jan-Willem, Silva Santos, Luiz Olavo Bonino da, Bourne, Philip E., Bouwman, Jildau, Brookes, Anthony J., Clark, Tim, Crosas, Mercè, Dillo, Ingrid, Dumon, Olivier, Edmunds, Scott C., Evelo, Chris T. A., Finkers, Richard, González-Beltrán, Alejandra N., Gray, Alasdair J. G., Groth, Paul, Goble, Carole A., Grethe, Jeffrey S., Heringa, Jaap, Hoen, Peter A. C. 't, Hooft, Rob W. W., Kuhn, Tobias, Kok, Ruben G., Kok, Joost N., Lusher, Scott J., Martone, Maryann E., Mons, Albert, Packer, Abel Laerte, Persson, Bengt, Rocca-Serra, Philippe, Roos, Marco, Schaik, Rene C. van, Sansone, Susanna-Assunta, Schultes, Erik Anthony, Sengstag, Thierry, Slater, Ted, Strawn, George O., Swertz, Morris A., Thompson, Mark, Lei, Johan van der, Mulligen, Erik M. van, Velterop, Jan, Waagmeester, Andra, Wittenburg, Peter, Wolstencroft, Katy, Zhao, Jun, and Mons, Barend, The FAIR Guiding Principles for scientific data management and stewardship, Scientific Data **3** (2016).

[Zhu+23] Zhu, Xixi, Liu, Bin, Zhu, Cheng, Ding, Zhaoyun, and Yao, Li, Approximate Reasoning for Large-Scale ABox in OWL DL Based on Neural-Symbolic Learning, Mathematics **11** no. 3 (2023).

Department of Mathematics, University of Colorado at Boulder, Boulder, CO
Email address: oscar-bender-stone@protonmail.com

DRAFT